

---

**TEST OF MATHEMATICS FOR  
UNIVERSITY ADMISSION****TMUA/01****Challenge Paper 1****75 minutes****Applications of Mathematical Knowledge****INSTRUCTIONS TO CANDIDATES**

Please read these instructions carefully, but do not open this question paper until you are told that you may do so.

A separate answer sheet is provided. Please check you have one. You also require a soft pencil and an eraser.

This paper contains twenty multiple-choice questions and should be attempted in 75 minutes. Each question has between six and ten options, labelled from A, exactly one of which is correct.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark. Record your choice for each question on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

You can use the question paper for rough working, but no extra paper is allowed. Only your responses on the answer sheet will be marked.

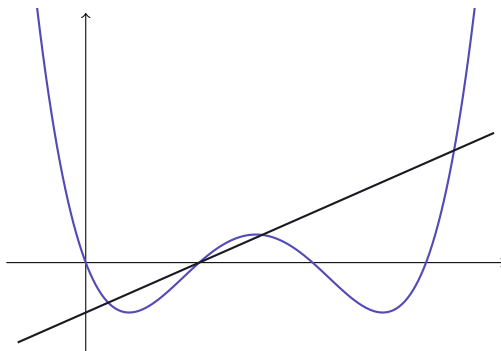
Calculators and dictionaries may NOT be used.

This is a Challenge Paper. It is set deliberately above the standard of the live test: the questions reward candidates who can combine several techniques and resist the obvious trap. Do not be discouraged by a low score.

Please wait to be told you may begin before turning this page.

This paper consists of twenty questions, a marked answer key, and full worked solutions.

- 1 The quartic  $y = x(x - 1)(x - 2)(x - 3)$  and the line  $y = x - 1$  meet at four points, as shown. Find the sum of the  $x$ -coordinates of these four points.



- A 3
- B 4
- C 6
- D 7
- E 10
- F 11
- G there are no such points
- H it cannot be found without solving the quartic

- 2 A sequence is defined by  $u_n = 4u_{n-1} - u_{n-2}$  with  $u_1 = 1$  and  $u_2 = 4$ . As  $k \rightarrow \infty$ , the ratio  $\frac{u_k}{u_{k-1}}$  tends to a limit. Find that limit.
- A 2
  - B 4
  - C  $2 + \sqrt{3}$
  - D  $2 - \sqrt{3}$
  - E  $\sqrt{3}$
  - F  $4 + \sqrt{3}$
  - G the ratio does not converge
  - H  $2\sqrt{3}$

- 3** A fair coin is tossed repeatedly until three heads appear in a row. Find the expected number of tosses.
- A** 6
  - B** 8
  - C** 12
  - D** 14
  - E** 16
  - F** 3
  - G** 7

4 Find the sum of the reciprocals of all the positive divisors of 360.

A 3

B  $\frac{13}{4}$

C  $\frac{39}{10}$

D 1170

E  $\frac{13}{12}$

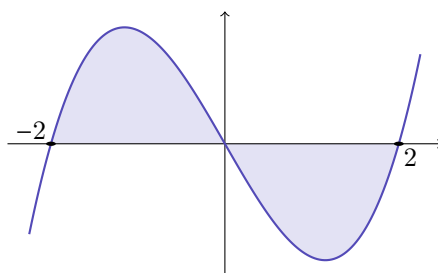
F  $\frac{9}{4}$

G  $\frac{117}{40}$

- 5** The digital root of a positive integer is obtained by summing its digits, then summing the digits of the result, and repeating until a single digit remains. What is the digital root of  $2^{2024}$ ?
- A** 1
  - B** 2
  - C** 4
  - D** 5
  - E** 7
  - F** 8
  - G** 9

- 6** How many real solutions does the equation  $2^{3x} - 7 \cdot 2^{2x} + 14 \cdot 2^x - 8 = 0$  have?
- A** 0
  - B** 1
  - C** 2
  - D** 3
  - E** 4
  - F** 6
  - G** infinitely many

- 7 The diagram shows the curve  $y = x^3 - 4x$ . Find the total area of the two shaded regions enclosed between the curve and the  $x$ -axis.



- A 0
- B 4
- C 8
- D 16
- E 12
- F  $\frac{32}{3}$
- G 24

**8** A function  $f$  satisfies  $f(x) + f(2 - x) = 4$  for every real  $x$ . Find  $\int_0^2 f(x) dx$ .

**A** 2

**B** 4

**C** 8

**D** 0

**E** it cannot be found without knowing  $f$

**F** 6

**G** 16

**9** Find the coefficient of  $x^2$  in the expansion of  $(1+x)(1+2x)(1+3x)\cdots(1+10x)$ .

- A** 55
- B** 220
- C** 385
- D** 1320
- E** 2640
- F** 3025
- G** 165

**10** How many solutions does  $\sin 3x = \sin x$  have in the interval  $0 \leq x < 2\pi$ ?

**A** 2

**B** 3

**C** 4

**D** 5

**E** 6

**F** 8

**G** infinitely many

**11** How many integers  $n$  with  $1 \leq n \leq 1000$  are divisible by neither 6 nor 10?

**A** 734

**B** 750

**C** 767

**D** 800

**E** 700

**F** 733

**G** 866

- 12** As  $x$  ranges over all real numbers, the expression  $\frac{x^2 + 2}{x^2 + x + 1}$  takes a greatest value. Find that greatest value.

**A** 2

**B**  $2 + \frac{2\sqrt{3}}{3}$

**C**  $2 - \frac{2\sqrt{3}}{3}$

**D** 3

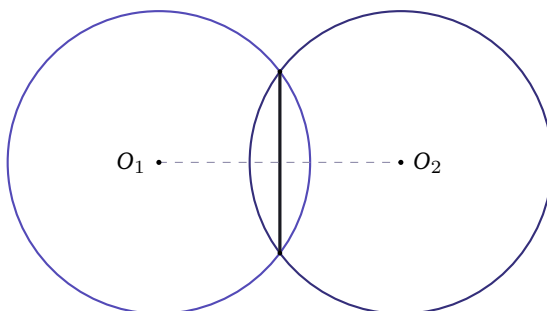
**E** 4

**F**  $2 + \sqrt{3}$

**G** there is no greatest value

**H**  $\frac{8}{3}$

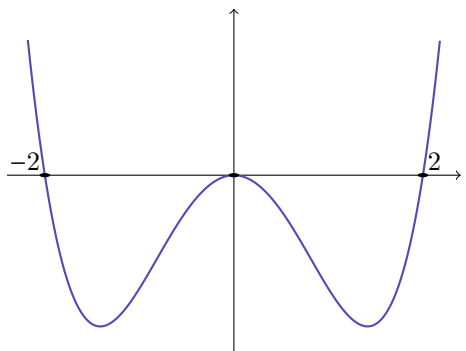
- 13** The diagram shows two circles,  $x^2 + y^2 = 25$  with centre  $O_1$  and  $(x - 8)^2 + y^2 = 25$  with centre  $O_2$ . They meet at two points. Find the length of their common chord.



- A** 3
- B** 6
- C** 8
- D**  $2\sqrt{3}$
- E**  $4\sqrt{3}$
- F**  $\sqrt{13}$
- G** the circles do not meet

- 14** A point  $P$  moves so that its distance from  $A(0, 0)$  is always twice its distance from  $B(6, 0)$ . The locus of  $P$  is a circle. Find its radius.
- A** 2
  - B** 3
  - C** 4
  - D** 6
  - E** 8
  - F** 12
  - G** the locus is not a circle

- 15** The diagram shows the curve  $y = x^4 - 4x^2$ . For how many values of the constant  $c$  does the horizontal line  $y = c$  meet the curve at exactly three points?



- A** 0
- B** 1
- C** 2
- D** 3
- E** 4
- F** infinitely many

- 16** The function  $f$  is defined for  $x \neq 1$  by  $f(x) = \frac{x+1}{x-1}$ . Find  $f^{2025}(x)$ , meaning  $f$  applied 2025 times.

**A**  $x$

**B**  $\frac{x+1}{x-1}$

**C**  $\frac{x-1}{x+1}$

**D**  $\frac{1}{x}$

**E**  $-x$

**F**  $\frac{1-x}{1+x}$

**17** Find the value of the infinite nested radical  $\sqrt{6 + \sqrt{6 + \sqrt{6 + \cdots}}}$ .

**A** 2

**B** 3

**C** 6

**D**  $\sqrt{6}$

**E**  $\frac{1 + \sqrt{5}}{2}$

**F** the expression does not converge

**G**  $\frac{1 + \sqrt{73}}{2}$

**18** Evaluate  $\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx$ .

**A** 0

**B**  $\frac{\pi}{6}$

**C**  $\frac{\pi}{4}$

**D**  $\frac{\pi}{3}$

**E**  $\frac{\pi}{2}$

**F** 1

**G**  $\ln 2$

- 19** In how many ways can three numbers be chosen from  $\{1, 2, 3, \dots, 20\}$  so that no two of them are consecutive? (Order does not matter.)
- A** 680
  - B** 816
  - C** 1140
  - D** 969
  - E** 1000
  - F** 560

- 20** For positive real numbers  $x$  and  $y$  with  $x + y = 1$ , find the least value of  $\left(1 + \frac{1}{x}\right)\left(1 + \frac{1}{y}\right)$ .
- A** 4
  - B** 6
  - C** 8
  - D** 9
  - E** 16
  - F** there is no least value
  - G** 3

## Answer key

---

| <b>Paper 1</b> |            |
|----------------|------------|
| <b>Q</b>       | <b>Ans</b> |
| 1              | C          |
| 2              | C          |
| 3              | D          |
| 4              | B          |
| 5              | C          |
| 6              | D          |
| 7              | C          |
| 8              | B          |
| 9              | D          |
| 10             | E          |
| 11             | C          |
| 12             | B          |
| 13             | B          |
| 14             | C          |
| 15             | B          |
| 16             | B          |
| 17             | B          |
| 18             | C          |
| 19             | B          |
| 20             | D          |

## Worked solutions

1 (C). The  $x$ -coordinates of the intersections are the roots of  $x(x-1)(x-2)(x-3) = x-1$ , that is of  $x^4 - 6x^3 + 11x^2 - 7x + 1 = 0$ . By Vieta, the sum of the roots of a quartic  $x^4 + px^3 + \dots$  is  $-p$ , here 6. Subtracting a linear function changes only the coefficients of  $x^1$  and  $x^0$ , never the  $x^3$  coefficient, so the sum is 6 whatever the line. The diagram confirms all four roots are real.

2 (C). The ratio of successive terms tends to the dominant root of the characteristic equation  $t^2 = 4t - 1$ , i.e.  $t^2 - 4t + 1 = 0$ , whose roots are  $t = 2 \pm \sqrt{3}$ . The larger,  $2 + \sqrt{3} \approx 3.73$ , dominates, so the ratio tends to  $2 + \sqrt{3}$ .

3 (D). Let  $E$  be the expected number of further tosses from a run of  $j$  heads. Standard first-step analysis gives, for a target run of length  $m$  with a fair coin,  $E = 2 + 4 + \dots + 2^m = 2^{m+1} - 2$ . For  $m = 3$  this is  $2^4 - 2 = 14$ .

4 (B). If  $\{d\}$  are the divisors of  $N$ , then  $\sum \frac{1}{d} = \frac{\sigma(N)}{N}$ , where  $\sigma$  is the sum of divisors. Here  $360 = 2^3 \cdot 3^2 \cdot 5$ , so  $\sigma(360) = (1 + 2 + 4 + 8)(1 + 3 + 9)(1 + 5) = 15 \cdot 13 \cdot 6 = 1170$ , and  $\frac{1170}{360} = \frac{13}{4}$ .

5 (C). The digital root of  $N$  equals  $N \bmod 9$  (taking 9 in place of 0). The powers of 2 modulo 9 cycle with period 6: 2, 4, 8, 7, 5, 1. Since  $2024 \equiv 2 \pmod{6}$ , we get  $2^{2024} \equiv 2^2 = 4 \pmod{9}$ , so the digital root is 4.

6 (D). Put  $y = 2^x > 0$ . The equation becomes  $y^3 - 7y^2 + 14y - 8 = 0$ , which factorises as  $(y-1)(y-2)(y-4) = 0$ . All three roots  $y = 1, 2, 4$  are positive, giving  $x = 0, 1, 2$ . Hence 3 real solutions.

7 (C). The curve crosses the axis at  $x = -2, 0, 2$ . By the symmetry  $f(-x) = -f(x)$  the two lobes have equal area. On  $[0, 2]$ ,  $\int_0^2 (4x - x^3) dx = \left[2x^2 - \frac{x^4}{4}\right]_0^2 = 8 - 4 = 4$ . The total shaded area is  $2 \times 4 = 8$ . (The plain integral  $\int_{-2}^2 f = 0$ , which is the trap.)

8 (B). Substituting  $x \mapsto 2 - x$  shows  $\int_0^2 f(x) dx = \int_0^2 f(2-x) dx$ . Adding,  $2 \int_0^2 f(x) dx = \int_0^2 [f(x) + f(2-x)] dx = \int_0^2 4 dx = 8$ , so the integral is 4. The particular  $f$  is never needed.

9 (D). The coefficient of  $x^2$  is the sum of all products  $i \cdot j$  with  $1 \leq i < j \leq 10$ , which equals  $\frac{1}{2}[(\sum i)^2 - \sum i^2] = \frac{1}{2}(55^2 - 385) = \frac{1}{2}(3025 - 385) = 1320$ .

10 (E).  $\sin 3x - \sin x = 2 \cos 2x \sin x = 0$ , so  $\sin x = 0$  or  $\cos 2x = 0$ . In  $[0, 2\pi)$ :  $\sin x = 0$  gives  $x = 0, \pi$ ;  $\cos 2x = 0$  gives  $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$ . That is 6 solutions in total.

11 (C). Among 1 to 1000: multiples of 6 number 166, multiples of 10 number 100, and multiples of  $\text{lcm}(6, 10) = 30$  number 33. By inclusion and exclusion, those divisible by 6 or 10 number  $166 + 100 - 33 = 233$ , so those divisible by neither number  $1000 - 233 = 767$ .

12 (B). Set  $y = \frac{x^2 + 2}{x^2 + x + 1}$  and clear denominators:  $(y-1)x^2 + yx + (y-2) = 0$ . For a real  $x$  the discriminant must be non-negative:  $y^2 - 4(y-1)(y-2) \geq 0$ , i.e.  $3y^2 - 12y + 8 \leq 0$ , giving  $2 - \frac{2\sqrt{3}}{3} \leq y \leq 2 + \frac{2\sqrt{3}}{3}$ . The greatest value is  $2 + \frac{2\sqrt{3}}{3} \approx 3.15$ .

13 (B). The two centres are 8 apart and both radii are 5. By symmetry the common chord is bisected perpendicularly by the line of centres, at distance 4 from each centre. Half the chord

is therefore  $\sqrt{5^2 - 4^2} = 3$ , so the chord has length 6. (Equivalently the circles meet where  $x = 4$ , giving  $y = \pm 3$ .)

14 (C). The condition  $PA = 2PB$  gives  $x^2 + y^2 = 4[(x-6)^2 + y^2]$ , which simplifies to  $x^2 + y^2 - 16x + 48 = 0$ , i.e.  $(x-8)^2 + y^2 = 16$ . This is a circle of radius 4 (an Apollonius circle).

15 (B). The curve has a local maximum at  $(0,0)$  and two minima at  $(\pm\sqrt{2}, -4)$ . A horizontal line  $y = c$  meets it at 4 points for  $-4 < c < 0$ , at 2 points for  $c > 0$  or  $c < -4$ , and at 2 points (tangentially) at  $c = -4$ . It meets the curve at exactly 3 points only at the local maximum value  $c = 0$ , where the roots are  $x = -2, 0, 2$ . So there is just 1 such value of  $c$ .

16 (B). Compute  $f(f(x))$ : with  $f(x) = \frac{x+1}{x-1}$ ,  $f(f(x)) = \frac{\frac{x+1}{x-1} + 1}{\frac{x+1}{x-1} - 1} = \frac{2x}{2} = x$ . So  $f$  is its own inverse and  $f^2$  is the identity. Applying  $f$  an odd number of times leaves one copy of  $f$ :  $f^{2025}(x) = f(x) = \frac{x+1}{x-1}$ .

17 (B). Let  $L$  be the value of the (increasing, bounded) nested radical. Then  $L = \sqrt{6+L}$ , so  $L^2 - L - 6 = 0$ , giving  $L = 3$  or  $L = -2$ . As  $L > 0$ , the value is 3.

18 (C). Let  $I$  be the integral. Replacing  $x$  by  $\frac{\pi}{2} - x$  swaps sine and cosine, giving  $I = \int_0^{\pi/2} \frac{\cos x}{\cos x + \sin x} dx$ . Adding the two forms,  $2I = \int_0^{\pi/2} 1 dx = \frac{\pi}{2}$ , so  $I = \frac{\pi}{4}$ .

19 (B). Choosing 3 items from a row of 20 with no two adjacent is equivalent to choosing 3 from  $20 - 3 + 1 = 18$  freely: the count is  $\binom{18}{3} = 816$ . (The trap  $\binom{20}{3} = 1140$  ignores the spacing condition.)

20 (D). Expand:  $(1 + \frac{1}{x})(1 + \frac{1}{y}) = \frac{(x+1)(y+1)}{xy} = \frac{xy + (x+y) + 1}{xy} = 1 + \frac{2}{xy}$ , using  $x+y=1$ . This is least when  $xy$  is greatest, i.e. at  $x=y=\frac{1}{2}$  giving  $xy = \frac{1}{4}$ , so the minimum is  $1 + \frac{2}{1/4} = 9$ .

